AI / MACHINE LEARNING INTERNSHIP PROJECT

Cancer Classification Using Neural Networks

An educational binary classifier for the Breast Cancer Wisconsin Diagnostic dataset.

Submitted by: _____

Institute / Department: _____

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CERTIFICATE

This is to certify that the project titled “Cancer Classification Using Neural Networks” has been prepared as part of the AI internship project work. The implementation, documentation and results described in this report are based on the supplied project data and the experiments performed for this submission package.

Student Signature: _____ Project Guide: _____

Date: _____ Institute Seal: _____

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I would like to express my sincere gratitude to the internship coordinator, project mentor and academic team for providing the opportunity and guidance to work on this project. I also acknowledge the developers of Python and the open-source machine-learning libraries used in this implementation. Finally, I thank my institution and peers for their support during the learning and development process.

ABSTRACT

An educational binary classifier for the Breast Cancer Wisconsin Diagnostic dataset. This report explains the problem, dataset, preprocessing, model design, implementation, evaluation and limitations. The system is intended for academic demonstration and learning. It should not be treated as a production system without additional validation, monitoring and domain review.

TABLE OF CONTENTS

1. 1. Introduction
2. 2. Problem Statement and Objectives
3. 3. Literature / Background Review
4. 4. Dataset Description
5. 5. Methodology
6. 6. System Design
7. 7. Implementation Details
8. 8. Experimental Setup
9. 9. Results and Discussion
10. 10. Testing and Validation
11. 11. Limitations and Risk Considerations
12. 12. Conclusion
13. 13. Future Scope
14. 14. References
15. Appendix A – How to Run

1. Introduction

Artificial intelligence and machine learning can identify patterns in structured data. In healthcare research, classification models are studied as decision-support prototypes, but they require rigorous validation before any clinical use.

This project implements a neural-network classifier using the Breast Cancer Wisconsin Diagnostic dataset. It predicts one of two dataset labels: malignant or benign.

The project is educational. Its purpose is to demonstrate data preparation, feature scaling, neural-network training and classification evaluation. It is not a clinical diagnostic system.

2. Problem Statement and Objectives

Problem statement: Use numerical measurements of cell nuclei from digitized breast-mass images to classify a record into the dataset's benign or malignant category.

Objectives:

- Understand the structure of the Wisconsin Diagnostic breast cancer dataset.
- Prepare features and target labels for supervised classification.
- Standardize numerical features using training-set statistics.
- Build and train a multilayer perceptron neural network.
- Evaluate performance using accuracy, precision, recall, F1 score and confusion matrix.
- Demonstrate a prediction on an unseen record while communicating medical limitations.

3. Literature / Background Review

A classification model learns a mapping from input features to discrete target classes. Binary classification has two possible output classes. Evaluation must consider more than accuracy when class errors have different consequences.

A feed-forward neural network consists of an input layer, one or more hidden layers and an output layer. Each neuron computes a weighted sum followed by an activation function. The ReLU activation is commonly used in hidden layers.

An MLP classifier is trained by backpropagation. The Adam optimizer updates weights using gradient information. Standardization is important because input variables with very different scales can make optimization difficult.

Precision measures the fraction of predicted positive cases that are correct. Recall measures the fraction of actual positive cases detected. F1 is the harmonic mean of precision and recall. A confusion matrix displays true and predicted class counts.

4. Dataset Description

The project uses the Breast Cancer Wisconsin Diagnostic dataset provided by scikit-learn. It contains 569 observations and 30 numerical features.

Features describe characteristics of cell nuclei, including radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry and fractal dimension. Measurements are represented by mean, standard error and worst-value feature groups.

The target encoding used by scikit-learn is 0 for malignant and 1 for benign. The dataset is a benchmark dataset and does not represent all patient populations, clinical settings or contemporary diagnostic workflows.

A copy of the feature table and target label is saved in data/breast_cancer_wisconsin.csv for transparency and inspection.

5. Methodology

First, load the dataset from scikit-learn and form a pandas DataFrame with feature names. Separate the 30 numerical predictors from the binary label.

Split the data into 80% training and 20% test/validation subsets. Stratified splitting preserves the approximate class distribution in both subsets. A fixed random seed of 42 makes the experiment reproducible.

StandardScaler transforms features to approximately zero mean and unit variance using statistics learned from the training subset. The scaler is placed inside a Pipeline to prevent the test subset from influencing preprocessing.

Train an MLPClassifier with two hidden layers containing 32 and 16 neurons, ReLU activation, Adam optimization and L2 regularization alpha=0.001. Early stopping monitors a validation fraction taken from the training data.

6. System Design

Input: 30 numerical measurements for each record.

Preprocessing: StandardScaler fitted on training observations; the same transformation is applied to unseen observations.

Neural network: feed-forward MLP with hidden layers (32, 16) and a binary output classification.

Evaluation: compare predicted labels with held-out labels using accuracy, precision, recall, F1 and confusion matrix. ROC AUC is also calculated from class probabilities.

Output: predicted label and class name. The model output is an educational estimate, not a medical conclusion.

7. Implementation Details

The implementation uses Python, pandas, NumPy and scikit-learn. The model is built as a Pipeline containing StandardScaler and MLPClassifier so that preprocessing and prediction remain connected.

The project guide's original notebook uses TensorFlow/Keras. The supplied implementation package provides an equivalent feed-forward neural-network approach using scikit-learn's MLPClassifier so it can run in a standard scikit-learn environment without TensorFlow installation.

The source script loads the benchmark dataset, splits it, fits the pipeline, prints evaluation metrics and demonstrates one held-out prediction. The notebook provides a step-by-step version of the same workflow.

The class mapping is explicitly printed in the README and code: 0 = malignant, 1 = benign. This avoids ambiguity when interpreting model outputs.

8. Experimental Setup

The experiment uses an 80:20 stratified split with random_state=42. The training subset is used for model fitting; the held-out subset is used for final validation metrics.

The MLP uses hidden_layer_sizes=(32,16), activation='relu', solver='adam', alpha=0.001, max_iter=500, early_stopping=True and n_iter_no_change=20.

The experiment reports accuracy, precision, recall, F1 score, confusion matrix and ROC AUC. The positive class for precision/recall/F1 is label 1 (benign) under the dataset's original encoding. The confusion matrix axes are labelled with the original class mapping.

9. Results and Discussion

The held-out validation accuracy is 0.9211 (92.11%). Precision is 1.0000, recall is 0.8750, and F1 score is 0.9333. ROC AUC is 0.9957.

The confusion matrix, with rows representing true labels and columns predicted labels in the order malignant (0), benign (1), is [[42, 0], [9, 63]].

The validation results show that the model learned useful patterns in this benchmark dataset. However, one split cannot establish clinical reliability. Results may change with different splits, populations, devices and data collection procedures.

Metric interpretation must consider both types of error. A false negative for malignant disease is particularly serious in a medical setting, which is one reason this student prototype must not be used for real diagnosis.

10. Testing and Validation

Dataset test: verify the dataset has 569 rows, 30 input features and no missing values in the supplied benchmark version.

Split test: use stratification and confirm that both classes appear in training and validation sets.

Preprocessing test: fit the scaler on training data only through the pipeline.

Model test: verify predictions are binary labels and class probabilities are finite.

Metric test: compute confusion matrix and classification metrics from the held-out data.

11. Limitations and Risk Considerations

This dataset is relatively small and curated. It may not represent demographic, geographic, clinical or technical diversity encountered in real healthcare systems.

A high benchmark score does not prove clinical validity, safety, fairness or generalizability. External validation and prospective clinical evaluation would be necessary before any real-world use.

The model does not include symptoms, patient history, imaging review by a clinician, pathology workflow or other clinical context. It should not be used to diagnose, rule out cancer or guide treatment.

The model's probability estimates may be poorly calibrated. Model explanations, bias assessment, privacy safeguards, governance and human oversight are not implemented in this academic prototype.

12. Conclusion

The project demonstrates a complete neural-network classification workflow on the Breast Cancer Wisconsin Diagnostic dataset. It includes dataset loading, stratified splitting, feature standardization, MLP model training and evaluation.

The validation experiment achieved 92.11% accuracy and an F1 score of 0.9333. The deliverables include source code, notebook, dataset copy, validation predictions, charts, report and presentation.

The result is useful for learning machine-learning concepts but is not evidence that the system is suitable for clinical use.

13. Future Scope

Evaluate the model using repeated stratified cross-validation and an independent external dataset.

Compare MLP performance with logistic regression, support vector machines, random forests and gradient boosting.

Study class-specific sensitivity and specificity, calibration, confidence intervals and decision thresholds.

Add explainability methods such as permutation importance or SHAP where appropriate, and document their limitations.

Any clinical research extension must involve qualified medical professionals, ethics review, privacy safeguards and rigorous validation.

APPENDIX A – HOW TO RUN

1. 1. Install Python 3.10 or later.
2. 2. Open a terminal in the project folder.
3. 3. Run: pip install -r requirements.txt
4. 4. Run the relevant script from src/ or open the notebook in Jupyter / Google Colab.
5. 5. Review the generated outputs in results/.
6. 6. Use the README for the complete repository workflow.

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APPENDIX B – DATA DICTIONARY

This appendix documents the meaning and role of the main fields used in the project. Input features describe each observation. The target field is the value the supervised model learns to predict. Data types and target encoding must be preserved when loading the data. Review the README and notebook for the exact column names and label mapping.

APPENDIX C – ALGORITHM WALKTHROUGH

The selected estimator learns patterns from labelled training observations. During fitting, model parameters are adjusted to reduce the training objective. Hyperparameters such as architecture, iteration count, learning rate and regularization influence model complexity and generalization. The chosen configuration is a reproducible baseline rather than a claim of globally optimal performance.

APPENDIX D – PREPROCESSING DETAILS

Preprocessing converts raw data into a representation suitable for a learning algorithm. Missing values are handled using statistics learned from training data. Categorical fields, where present, are encoded into model-readable indicators. Numerical scaling is used in the neural-network project. All fitted transformations are applied consistently to validation and future records.

APPENDIX E – TEST CASES

Test 1: load the supplied or documented dataset and verify its dimensions. Test 2: inspect target labels and feature columns. Test 3: perform the train/validation split with the documented random seed. Test 4: fit preprocessing and the estimator using training data only. Test 5: generate predictions for held-out rows. Test 6: calculate the metrics and confirm that output files can be reopened.

APPENDIX F – RESULT INTERPRETATION

Evaluation metrics summarize performance on the held-out validation sample. They are not guarantees of future performance. Always interpret a metric with its definition, units, class mapping and validation design. Visualizations should be read alongside the numeric metrics, not as substitutes for them. The results folder contains the plots produced by the experiment.

APPENDIX G – GLOSSARY

Feature: an input variable. Target: the value to be predicted. Training set: observations used to fit the model. Validation set: held-out observations used to estimate generalization. Imputation: replacement of missing values. Overfitting: learning training-specific patterns that do not generalize. Reproducibility: ability to repeat an experiment using recorded data, code and settings.

APPENDIX H – REPRODUCIBILITY AND REPOSITORY

Keep data files, scripts, notebooks, reports, presentation and results in the documented folder structure. Install packages from requirements.txt. Run the Python script or execute the notebook from top to bottom. Keep the fixed random seed unchanged when reproducing the documented metrics. Do not upload virtual environments, cache files or private credentials to GitHub.

APPENDIX I – MEDICAL SAFETY AND ETHICS

This classifier is an educational benchmark exercise only. It has not been clinically validated and must not be used to diagnose, screen, rule out cancer or guide treatment. Dataset performance does not establish performance across hospitals, populations, scanners or clinical workflows. Any research extension involving patient data requires appropriate authorization, privacy protections, ethics oversight, qualified medical supervision and independent validation.